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Perspective

Quantifying entanglement from the geometric perspective

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Abstract – Quantum entanglement between several particles is essential for applications like quantum metrology or quantum cryptography, but it is also central for foundational phenomena like quantum non-locality. This leads to the problem of quantifying the amount of entanglement in a quantum state. We present a review on the geometric measure of entanglement, being a quantifier based on the distance of a state to the nearest separable state. We explain basic properties, existing methods to compute it, its operational interpretations, as well as scaling and complexity issues. We point out intimate relations to fundamental problems in mathematics concerning eigenvalues and norms of tensors. Consequently, the geometric measure of entanglement provides a playground where physical intuition and mathematical rigor benefit from each other.




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Introduction. – Entanglement plays an important role in quantum information theory and is often considered to be a resource for quantum metrology, quantum cryptography or other applications [1,2]. As such, there is an ongoing search for characterization and quantification techniques, measuring the amount and usefulness of entanglement in quantum states [3,4]. Here, different use cases lead to different approaches on entanglement quantification, resulting in different measures of entanglement. One approach for quantifying entanglement is given by the geometric measure of entanglement, which quantifies the entanglement of a state by its distance to the separable states. This entanglement measure was first introduced by Shimony for bipartite pure states [5], extended to multipartite pure states by Barnum and Linden [6] and fully generalized to mixed states by Wei and Goldbart [7].

A remarkable and important feature of the geometric measure is its close relation to different concepts from mathematics. Not only is the geometric measure directly linked to the injective tensor norm, it is also related to the theory of tensor eigenvalues and tensor decompositions. In this article we wish to point out different connections between the physical and mathematical notions, hopefully leading to a better understanding of the interplay of physics and mathematics.

Basic definitions. – We start with the simple case of a pure state $|\psi\rangle = \sum_{ij} \tau_{ij} |i\rangle_A \otimes |j\rangle_B$ of two particles, each described as a d -dimensional system. Here, the complex coefficients τ_{ij} form a $d \times d$ matrix. If this state can be written as

$$|\psi\rangle = |a\rangle_A \otimes |b\rangle_B \equiv |ab\rangle, \quad (1)$$

each subsystem can independently be described by the respective d -dimensional vectors $|a\rangle_A$ and $|b\rangle_B$. Consequently, the state $|\psi\rangle$ is called separable. If this is not the case, the state $|\psi\rangle$ is called entangled.

For checking the separability of $|\psi\rangle$ note that every state can be written in the Schmidt decomposition

$$|\psi\rangle = \sum_{i=1}^r s_i |a_i b_i\rangle, \quad (2)$$

where the real Schmidt coefficients s_i are positive, unique, fulfill the normalization $\sum_{i=1}^r s_i^2 = 1$ and are typically ordered as $s_1 \geq s_2 \geq \dots$. The Schmidt vectors $|a_i\rangle$ (and $|b_i\rangle$) are orthonormal and are also unique, if the Schmidt coefficients are pairwise different. The number r is the so-called Schmidt rank of the state. The Schmidt decomposition follows directly from the singular value decomposition of the complex coefficient matrix τ_{ij} . Clearly, a state is separable, if and only if the Schmidt rank equals one, $r = 1$; or, in other words, if the matrix τ_{ij} has rank one. The states with maximal Schmidt rank $r = d$ and

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identical Schmidt coefficients $s_i = 1/\sqrt{d}$ are the generalized Bell states and maximally entangled.

Given an entangled state, one can naturally consider its distance to the set of all separable states as a quantifier of the amount of entanglement. In two-particle systems this maximal overlap $\Lambda(\psi)$ can be easily calculated by using the Schmidt decomposition, resulting in

$$\Lambda(\psi) := \max_{|ab\rangle} |\langle ab|\psi\rangle| = s_1. \quad (3)$$

In the mathematical literature, this is known as the variational characterization of the Ky-Fan norm of τ_{ij} [8]. One then defines the geometric measure of entanglement [5–7]

$$E(\psi) = 1 - \Lambda^2(\psi). \quad (4)$$

Clearly, this measure vanishes for all separable states, while the maximum is given by $E = 1 - 1/d$.

Naturally, one would like to apply this approach also to multipartite states. We consider as an example a tripartite state $|\psi\rangle = \sum_{ijk} \tau_{ijk} |ijk\rangle$, but the following definitions can directly be generalized to N parties. The maximal overlap for three parties is, as in the bipartite case,

$$\Lambda(\psi) = \max_{|abc\rangle} |\langle abc|\psi\rangle| \quad (5)$$

and the geometric measure $E(\psi) = 1 - \Lambda^2(\psi)$.

In practice, however, a significant difference from the bipartite case arises: The coefficients are given by a tensor τ_{ijk} , and no easy-to-calculate singular value decomposition for tensors exists. Consequently, there is no Schmidt decomposition in the multipartite case (see also the discussion after eq. (15) below), complicating the calculation of the geometric measure considerably.

As examples, consider for three qubits the Greenberger-Horne-Zeilinger (GHZ) state and the W state,

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}} (|000\rangle + |111\rangle), \quad (6)$$

$$|W\rangle = \frac{1}{\sqrt{3}} (|001\rangle + |010\rangle + |100\rangle). \quad (7)$$

For the GHZ state, we can resort to the bipartite case by noticing that the state is already in the Schmidt decomposition for the bipartition $A|BC$ with Schmidt coefficients $s_1 = s_2 = 1/\sqrt{2}$. Consequently, the maximal overlap with biseparable states of the type $|\eta\rangle = |a\rangle_A \otimes |\phi\rangle_{BC}$ is given by $1/\sqrt{2}$. Since a product state $|abc\rangle$ is a special case of the state $|\eta\rangle$, this is also an upper bound on $\Lambda(\text{GHZ})$. In fact, the product state $|000\rangle$ reaches exactly this value, leading to $\Lambda^2(\text{GHZ}) = 1/2$ and $E(\text{GHZ}) = 1/2$.

For the W state, this method gives only an upper bound. But, as we will explain in the next section, we can use the fact that for a permutationally symmetric state the maximizing product state is also symmetric, *i.e.*, one can restrict the attention to product states of the form $|aaa\rangle$. Using this, one finds $\Lambda^2(W) = 4/9$ and $E(W) = 5/9$, so for the geometric measure, the W state is more entangled than the GHZ state.

Basic results. – Let us start by discussing some methods to calculate or estimate the geometric measure. First, a simple numerical approach to calculate $\Lambda(\psi)$ can be formulated as a see-saw algorithm [9,10]. This relies on the fact that if all but one part of the product state are fixed, then the optimal remaining part can directly be found. More precisely, for given fixed states $|b_0\rangle$ and $|c_0\rangle$ one can calculate the unnormalized state $|\alpha\rangle = \langle b_0 c_0 | \psi \rangle$ on Alice's space. Then, it is clear that the overlap $|\langle ab_0 c_0 | \psi \rangle|$ is maximal, if we choose $|a\rangle$ proportional to $|\alpha\rangle$, that is $|a\rangle \propto |\alpha\rangle$. This can be extended to an iteration, where one starts with a random product state $|a_0 b_0 c_0\rangle$ and then updates first Alice's, then Bob's and then Charlie's vector, and then starts with Alice again, etc. Repeating this leads to a monotonically increasing sequence of values and hence to a fixed point. This is often (but not always) the global optimum, in practice it works well for states up to about seven qubits [9–11]. In fact, the fixed points are the so-called singular values of the tensor τ_{ijk} , which are related to Z-eigenvalues of tensors, see also below.

Strictly speaking, the iteration above provides only a lower bound on the value of $\Lambda(\psi)$. Upper bounds can be derived considering a relaxation of the problem using the separability criterion of the positivity of the partial transposition (PPT) [12,13]. For that, note that instead of optimizing the overlap with all pure separable states, one can define alternatively $\Lambda^2(\psi) = \max_{\sigma \in \text{f.s.}} \text{Tr}(|\psi\rangle\langle\psi|\sigma)$ where the optimization runs over all fully separable states of the form $\sigma = \sum_k p_k |a_k b_k c_k\rangle\langle a_k b_k c_k|$ [2]. The set of fully separable states is strictly smaller than the set of states obeying the PPT criterion for each bipartition, that is, $F_{\text{PPT}} = \{\rho, \rho^{Tx} \geq 0, X = A, B, C\}$. So,

$$\Lambda^2(\psi) \leq \max_{\rho \in F_{\text{PPT}}} \text{Tr}(|\psi\rangle\langle\psi|\rho) \quad (8)$$

denotes an upper bound. Importantly, the optimization over fully PPT states can be formulated as a semidefinite program, which can be solved efficiently, see ref. [14] for a recent concrete example. The bound (8) is better than the upper bound from considering all bipartite Schmidt decompositions of the state $|\psi\rangle$ (as outlined above for the GHZ state), which follows from connections between quantum state fidelities and the PPT criterion [15,16].

Often the computation of the geometric measure can be simplified if $|\psi\rangle$ has specific properties. As a simple example, if $|\psi\rangle$ has only real and positive coefficients, then the product state can be assumed to have the same structure, see, *e.g.*, [17]. A highly relevant case arises if the state $|\psi\rangle$ is symmetric, *i.e.*, if the state is invariant under the permutation $P_{\alpha,\beta}$ of any two particles α and β , $|\psi\rangle = P_{\alpha,\beta}|\psi\rangle$. Symmetric states of N parties of dimension d form a subspace of dimension $\binom{N+d-1}{d-1}$ and the entanglement theory in this space is well developed, see, *e.g.*, refs. [18–21].

For example, the GHZ and the W state are symmetric states. It seems rather natural that the maximal over-

lap $\Lambda(\psi)$ with a symmetric state should be attainable by a symmetric product state, that is, a state of the form $|aaa\rangle$. Indeed, this was known already in mathematics for some time in the context of symmetric polynomials over Banach spaces [22], see also the discussion in [23]. In fact this result was then extended by proving that, in the case of $N \geq 3$ parties, the closest product state is necessarily symmetric [23]. Note, however, that not every structure of the state $|\psi\rangle$ can directly be also assumed for the closest product state. A simple example is the translationally invariant state $|\psi\rangle = (|0101\rangle + |1010\rangle)/\sqrt{2}$, for which $\Lambda^2(\psi) = 1/2$ is attained for the product states $|0101\rangle$ and $|1010\rangle$, which are not translationally invariant [23].

A direct consequence of these results is that for the W state $\Lambda^2(W) = 4/9$, as explained above. In addition, the geometric measure is also known for other families of states, such as Dicke states and other symmetric states [7,24,25], or graph and hypergraph states [17,26,27].

The discussed methods can be used to obtain structural insights into the geometric measure. First, the see-saw algorithm can be extended (and used as a subroutine) to search for maximally entangled states [11]. The main idea is the following: Consider a state $|\psi\rangle$ and the closest product state $|abc\rangle$, potentially computed by the see-saw algorithm, and the updated state $|\phi\rangle = (|\psi\rangle + \varepsilon|\eta\rangle)/\mathcal{N}$, where $|\eta\rangle \propto [1 - |abc\rangle\langle abc|]|\psi\rangle$ is chosen orthogonal to $|abc\rangle$ and $\mathcal{N}$ denotes a normalization. It can be shown that if the closest product state $|abc\rangle$ is unique and ε is sufficiently small, then the geometric measure of $|\phi\rangle$ is strictly larger than the one of $|\psi\rangle$. Iterating this procedure leads to states with maximal geometric measure, for example, for three qubits one finds the W state. This works well up to six qubits, but also in higher dimensions [11].

A variation of this algorithm can be used to search for highly entangled subspaces. Interestingly, for three qubits one finds a two-dimensional subspace spanned by the W state in eq. (7) and

$$|V\rangle = \frac{1}{\sqrt{3}} \left(|011\rangle + e^{\frac{2\pi}{3}i} |110\rangle + e^{\frac{4\pi}{3}i} |101\rangle \right). \quad (9)$$

All states in this subspace have the maximally reachable geometric measure for three qubits of $5/9$, similar constructions can be generalized [28].

The results about symmetric pure states, moreover, can be used to show the non-multiplicativity of $\Lambda(\psi)$, often also called non-additivity, if one considers the logarithm $\hat{E}(\psi) = -\log[\Lambda(\psi)]$ as a variant of the geometric measure. For that, one takes two N -particle states $|\psi\rangle$ and $|\phi\rangle$ and considers the combined state $|\psi\rangle \otimes |\phi\rangle$ as an N -partite state, but with higher local dimension. Then, in general it does not hold that $\Lambda(\psi \otimes \phi) = \Lambda(\psi)\Lambda(\phi)$. Indeed, considering a completely antisymmetric state $|\psi\rangle_{\text{as}}$ (meaning that $P_{\alpha,\beta}|\psi\rangle = -|\psi\rangle$ for all permutations) we find that the closest product state cannot be symmetric, as otherwise their overlap would vanish. However, the state $|\psi\rangle_{\text{as}} \otimes |\psi\rangle_{\text{as}}$ is *symmetric* on the N -partite system, so the nearest product state in this higher-dimensional space

has to be symmetric and can therefore not consist of two copies of the closest product state to $|\psi\rangle_{\text{as}}$ [29,30]. This non-additivity is related to the additivity of the output purity of quantum channels [29,31,32], see also below.

Operational interpretations. – The geometric measure has an operational interpretation in the context of state discrimination. Considering K orthogonal states $\{|\psi_i\rangle, i = 1, \dots, K\}$ one can ask whether one can discriminate them perfectly. While this is clearly possible if global measurements are allowed, the question is more interesting and more relevant if one allows only local operations and classical communication (LOCC). In fact, the maximal number of states that can be discriminated by LOCC is upper bounded by their geometric measure [33].

To see this, note that perfect state discrimination of the states $\{|\psi_i\rangle\}$ is only possible if there exists a generalized measurement with the effects $\{M_i\}$, such that each effect detects one of the states perfectly, that is $\text{Tr}(M_i|\psi_i\rangle\langle\psi_i|) = 1$. If the parties are only allowed to perform LOCC operations, the effects M_i additionally have to be separable. Noting that the effects correspond to density matrices ω_i up to some scaling, optimizing the measurement for the given state set reduces to an optimization over separable density matrices, similar to the optimization over product states in the geometric measure. More precisely, one finds that if the discrimination of the states $\{|\psi_i\rangle, i = 1, \dots, K\}$ in a d^N -dimensional space is possible, then $\sum_{i=1}^K 1/\Lambda^2(\psi_i) \leq d^N$ holds. Since separable states have $\Lambda^2(\psi_i) = 1$, one finds that d^N separable states forming a basis can potentially be discriminated (although this does not need to be possible [34]), but choosing one of the states to be entangled (*i.e.*, $\Lambda^2(\psi_i) < 1$), perfect discrimination by LOCC is no longer possible. More generally, the more entangled the states are in terms of the geometric measure, the fewer states can be perfectly discriminated.

There are further operational interpretations, *e.g.*, in the context of measurement-based quantum computation or in relation to quantum channels, which we will explain below. Finally, the geometric measure has been used to characterize phase transitions in condensed matter systems, see refs. [35–39] for examples.

Asymptotic behaviour. – Having reviewed properties of the geometric measure for specific states, the question arises whether statistical statements for general states can be made, especially for many particles.

A simple argument supports the conjecture that generic states of many particles are highly entangled: The description of a generic state of N particles requires an exponentially increasing number of parameters (in N), while for product states this number increases only linearly. So, the fraction of product states decreases with N and a generic state can be expected to be highly entangled. In fact, it was already shown in ref. [40] that for a generic multiparticle state (distributed according to the Haar measure) the probability that there is *some* bipartite cut, for which the

state is *not* highly entangled (measured by the entropy of the reduced state) is exponentially suppressed.

Concerning the geometric measure for multiqubit states, it was shown [41] that for $N > 11$ the probability of finding a state with small geometric measure is constrained via

$$\text{Prob}\{\Lambda^2(\psi) > 3N^2 \times 2^{-N}\} < \exp\{-N^2\}. \quad (10)$$

Note that expressing a normalized state in the computational basis yields for any state the bound $\Lambda^2(\psi) \geq 2^{-N}$, so eq. (10) implies that generic multi-qubit states have a geometric measure close to the maximally possible one.

The proof idea is based on two facts. First, it is well known that if one takes a fixed vector $|\phi\rangle$ and a random vector $|\psi\rangle$ in a high-dimensional space, then their overlap is with high probability close to zero. Second, one can approximate product states with a so-called ε -net. This is a small set of states $\{|\phi_i\rangle\}$, such that for any product state $|\pi\rangle$ one finds a state in the ε -net which approximates $|\pi\rangle$ well. The ε -net for N qubits can be constructed from approximations of the Bloch sphere for a single qubit.

These facts can be combined as follows. For a random state $|\psi\rangle$ one asks for the probability that the maximum of the overlaps with states in the ε -net is high. Using a union bound, this can be upper bounded by the product of the number of states in the ε -net and the probability for a fixed state $|\phi_i\rangle$ to have a high overlap with $|\psi\rangle$. Due to the first fact, the latter is also bounded, leading to the conclusion that the probability of $|\psi\rangle$ having a good approximation in the ε -net is small. Since the ε -net is a good approximation of the product states, this also holds if one wishes to find the maximal overlap of $|\psi\rangle$ with product states.

This fact has direct consequences for the usefulness of generic states in measurement-based quantum computation [42], where one prepares an entangled state on many qubits, for instance the so-called cluster state, and then implements an arbitrary quantum algorithm by performing local measurements. Naturally, this requires strong correlations between the outcomes of the single qubits. The result in eq. (10) implies, however, that for a generic state the measurement results in any local basis are highly uncorrelated [41,43]. Hence, generic states cannot be used for measurement-based quantum computation or, in the words of ref. [41], they are *too entangled to be useful*.

We add that further scaling results of the geometric measure have been derived, for instance if states have a permutational symmetry [44], for higher dimensions [30], as well as upper bounds obtainable for fixed dimensions [45]. An interesting open question is whether there is a constant C such that for any N there is a multi-qubit state for which $\Lambda^2(\psi) \leq C \times 2^{-N}$ (see Problem 8.27 in ref. [46]). More discussions, also about the distribution of $\Lambda^2(\psi)$ for small N can be found in refs. [11,47].

Mathematical concepts. – Here we discuss related concepts from multilinear algebra and tensor analysis.

Injective tensor norm. First, the quantity $\|\tau\| = \sup_{\|x\|=\|y\|=\|z\|=1} |\sum_{ijk} \tau_{ijk} x_i y_j z_k|$, equaling $\Lambda(\psi)$ in our

notation, is also called the injective tensor norm, and several other problems have been shown to be related to its computation. For instance, considering a quantum channel $S(\rho) = \sum_k A_k \rho A_k^\dagger$ in the Kraus representation, one may be interested in the maximal purity of the output of this channel. For that, one can consider the quantity $\mathcal{P} = \langle \eta | S(|\phi\rangle\langle\phi|) | \eta \rangle$, which can be calculated as $\mathcal{P} = \sum_k |\langle \eta | A_k | \phi \rangle|^2 = \sup_{\|x\|=1} |\sum_k x_k \langle \eta | A_k | \phi \rangle|^2 = \sup_{\|x\|=1} |\sum_{ijk} x_k \eta_i^* A_k^{ij} \phi_j|^2$, where the indices i, j label the entries of the matrices A_k . Thus, $\sup_{|\eta\rangle, |\phi\rangle} \mathcal{P} = \|A_k^{ij}\|^2$ holds, relating the additivity of the output purity of channels to the additivity of the injective tensor norm [29].

Moreover, the complexity of computing the injective tensor norm can be directly related to the complexity of the bipartite separability problem for mixed states. For that, note that the basic idea of the iterative algorithm for computing $\Lambda^2(\psi)$ can be rephrased as follows: If we define the bipartite operator $X_{AB} = \text{Tr}_C(|\psi\rangle\langle\psi|)$ on the first two subsystems, then

$$\Lambda^2(\psi) = \sup_{|a\rangle, |b\rangle} \langle ab | X_{AB} | ab \rangle. \quad (11)$$

In other words, the maximal overlap of a bipartite operator X_{AB} with product states equals the parameter $\Lambda^2(\psi)$ of its purification. The optimization in eq. (11) is essentially equivalent to the construction of bipartite entanglement witnesses, allowing to translate complexity results from the bipartite separability problem to the injective tensor norm on higher-order tensors, see ref. [48] for a detailed discussion. Finally, tensors with maximal injective tensor norm are cases where the solution of the MAX-N-local Hamiltonian problem, where one asks for the eigenvector to the largest eigenvalue of a Hamiltonian, and its best product approximation deviate maximally [49].

Tensor eigenvalues. Another related concept are eigenvalues of tensors. For their study, typically real and symmetric tensors are considered [50]. The reason is that one is interested in the positivity of polynomials of the type $f(x) = \sum \tau_{ijkl} x_i x_j x_k x_l = \langle \psi | x x x x \rangle$ for real tensors τ_{ijkl} , and such a polynomial does not capture non-symmetric properties of the tensor. Additionally, for a given non-symmetric tensor one can also always consider a symmetric embedding into a larger symmetric tensor [51]. We here focus our discussion on tensors with four indices, but anything can directly be generalized to N indices.

For real symmetric tensors, there are two important classes of tensor eigenvalues. The first class are the H-eigenvalues, which are defined by

$$\langle \psi | x x x \rangle = \lambda |x^3| \Leftrightarrow \sum_{jkl} \tau_{ijkl} x_j x_k x_l = \lambda x_i^3, \quad (12)$$

where $\lambda \in \mathbb{R}$ is the real eigenvalue, $|x\rangle \neq 0$ is the real eigenvector and $|x^3\rangle$ denotes the vector with the components x_i^3 . The seemingly strange power of 3 (or, more generally, $N-1$) on the RHS assures for a given λ that every (scaled) vector $\alpha|x\rangle$ is an eigenvector, too. The H-eigenvalues fulfill

many of the properties of matrix eigenvalues. For instance, the H-eigenvalues of a real tensor for even N are exactly the diagonal elements, there exists a Gershgorin-disc-type theorem and the product of the eigenvalues equals $\det(\tau)$, where $\det(\cdot)$ is the hyperdeterminant used to calculate the eigenvalues [50].

The second class are Z-eigenvalues, defined by

$$\langle \psi | xxx \rangle = \lambda \langle x | \text{ with } \langle x | x \rangle = 1, \quad (13)$$

where again $\lambda \in \mathbb{R}$ is the real eigenvalue and $|x\rangle \neq 0$ is the corresponding eigenvector. These eigenvalues are not directly comparable to matrix eigenvalues, as scaling a Z-eigenvector $\alpha|x\rangle$ leads in general not to an Z-eigenvector again. Additionally, the Z-eigenvalues are not necessarily given by the diagonal elements of diagonal tensors. As an example, consider the (unnormalized) state $|\psi\rangle = 3|0000\rangle + |1111\rangle$, where the coefficient tensor τ is diagonal with diagonal elements $\tau_{0000} = 3$ and $\tau_{1111} = 1$. Besides the two obvious Z-eigenpairs $|x^{(1)}\rangle = |0\rangle$ for $\lambda^{(1)} = 3$, and $|x^{(2)}\rangle = |1\rangle$ for $\lambda^{(2)} = 1$, there exists the third Z-eigenpair $|x^{(3)}\rangle = 1/2(|0\rangle + \sqrt{3}|1\rangle)$ for $\lambda^{(3)} = 3/4$ [50]. This property is important when calculating the eigenvectors or the geometric measure by a see-saw algorithm, as every eigenvector is a fixed point of the iteration. On the other hand, Z-eigenvalues have the nice property of being invariant under any local orthogonal transformation $|\psi\rangle \mapsto O \otimes O \otimes O |\psi\rangle$. Both classes of eigenvalues are related to the positivity of the polynomial $f(x)$ for even N , as the polynomial is positive, if and only if all H-eigenvalues (or, equivalently, all Z-eigenvalues) are positive [50].

The above notions can be generalized to complex tensors, complex eigenvalues or complex eigenvectors, leading to eigenvalues denoted by different letters. For example, E-eigenvalues can be obtained from the definition of Z-eigenvalues as in eq. (13) by allowing for complex eigenvalues $\lambda \in \mathbb{C}$ and eigenvectors $|x\rangle \in \mathbb{C}^d$, but still considering the Euclidean normalization condition $\sum_i x_i^2 = \langle x^* | x \rangle = 1$. This is not the optimization leading to $\Lambda^2(\psi)$ for complex quantum states, as the normalization condition is relevant. For example, for $|\psi\rangle = |000000\rangle + |111111\rangle$, the state $|x\rangle = 1/\sqrt{2}(|0\rangle + i|1\rangle)$ fulfills the first condition for an E-eigenvalue, but its Euclidean norm is $\sum_i x_i^2 = \langle x^* | x \rangle = 0$, so it is not normalizable.

Another relevant class are the US-eigenvalues of complex symmetric tensors, which are directly related to the geometric measure [52]. These are defined as in eq. (13) where λ is the real eigenvalue and $|x\rangle$ is the according (complex) eigenvector with the standard normalization $\sum_i |x_i|^2 = \langle x | x \rangle = 1$. This definition directly yields the geometric measure of a symmetric state as the maximal US-eigenvalue. If the state is in addition real and non-negative, the maximal US-eigenvalue reduces to the largest Z-eigenvalue [53].

Concerning the computation of eigenvalues, we would like to add two more facts. First, for finding the largest US-eigenvalue one may consider the set $\mathcal{X} = \{\|x\| \mid \langle \psi | (|x\rangle^{\otimes(N-1)}) = \langle x |, |x\rangle \neq 0\}$. Then, $\max\{\lambda\} =$

$(1/\min \mathcal{X})^{N-2}$, transforming the problem into a system of algebraic equations [52]. A different approach to calculate the different eigenvalues is the so-called tensor power method, being similar to the see-saw algorithm described above. In this method, one starts from a symmetric state $|x^{(0)}x^{(0)}x^{(0)}x^{(0)}\rangle$, and, after calculating the overlap $\langle \psi | x^{(0)}x^{(0)}x^{(0)}x^{(0)}\rangle$, updates every part of the vector according to $|x^{(1)}\rangle \propto \langle \psi | x^{(0)}x^{(0)}x^{(0)}x^{(0)}\rangle$.

Finally, one can also define singular values for tensors. For that, one considers an entire set $\{|x^{(1)}\rangle, \dots, |x^{(N)}\rangle\}$ of vectors which obey equations like

$$\langle \psi | x^{(1)}, \dots, \bar{x}^{(k)}, \dots, x^{(N)} \rangle = \lambda \langle x^{(k)} |, \quad \langle x^{(i)} | x^{(i)} \rangle = 1 \quad \forall i. \quad (14)$$

The notation $\bar{x}^{(k)}$ denotes that the vector $|x^{(k)}\rangle$ is missing in the tensor product. Naturally, this is closely related to the notion of entanglement eigenvalue equations, which have been used to analyze the geometric measure [7,10].

We would like to add that tensor eigenvalues have indeed been used in quantum physics besides the notions of the geometric measure. In ref. [54] the authors expressed symmetric multi-qubit states in the Bloch representation, *i.e.*, in terms of expectation values of tensor products of Pauli matrices. This leads to a real, symmetric tensor, and it is then shown that if the smallest Z-eigenvalue is negative, the state is entangled.

Odeco and fradeco decompositions. The concepts of the geometric measure, the injective tensor norm as well as tensor eigenvalues aim at approximating a tensor τ_{ijk} or specific properties of it by a tensor of the form $\eta_{ijk} = a_i b_j c_k$, which may be considered as a rank-one tensor. It is natural to ask for further mathematical concepts to define extended notions of correlations in quantum information.

A first question concerns the generalization of the Schmidt decomposition to three or more particles, asking whether a state can be written as

$$|\psi\rangle = \sum_{i=1}^r s_i |a_i b_i c_i\rangle, \quad (15)$$

where the $|a_i\rangle$ ($|b_i\rangle$, $|c_i\rangle$) form an orthonormal basis of the respective local Hilbert space. A simple argument shows that this is in general not possible [55]: A decomposition as in eq. (15) constitutes also a bipartite Schmidt decomposition as in eq. (2) for the partition $A|BC$, but with the additional constraint that the Schmidt vectors on the joint party BC are product vectors. Since the bipartite Schmidt decomposition in eq. (2) is unique, one cannot expect this condition to be met for generic $|\psi\rangle$. Still, normal forms of multiparticle states have been derived [56,57].

A decomposition as in eq. (15) is possible for a specific class of tensors, and in the mathematical literature real and symmetric tensors with this property are called orthogonally decomposable (odeco) [58]. For these tensors, an explicit form of all Z-eigenvectors in terms of the Schmidt vectors in eq. (15) can be found [58]. Moreover,

for odeco tensors the Schmidt vectors are exactly the robust fixed points of the tensor power iteration outlined above, so that they can be found numerically [58,59].

A potential generalization of odeco tensors are tensors that can be decomposed into frames (fradeco), *i.e.*, tensors where a decomposition as in eq. (15) can be found and the $|a_i\rangle$ (and $|b_i\rangle$ and $|c_i\rangle$) form a frame instead of a basis [60]. A (unit-norm) frame is a set of r (normalized) vectors $\{|v_k\rangle\}$ that form an overcomplete basis in the sense that for any vector $|w\rangle$ one has $A\|w\|^2 \leq \sum_{i=1}^r |\langle w|v_k\rangle|^2 \leq B\|w\|^2$ with some constants A and B , and the frame is tight if $A = B$ (for an introduction on frames see, *e.g.*, [61]). For fradeco tensors, the tensor power method for calculating eigenvalues does not always find all eigenvectors, as some of them belong to repelling fixed points [62].

More generally, one can ask how many terms in a decomposition like eq. (15) are needed, even if no restriction on the local vectors $|a_i\rangle$ etc. is made. Such decompositions can be represented graphically [63], and can be seen as dual to the injective tensor norm [45]. The resulting minimal $r_{\min}$ is called the tensor rank of the tensor τ_{ijk} . This is central for many applications, such as the efficient numerical matrix multiplication [64]. Its computation, however, is difficult as it depends on the underlying field (a real tensor may have a different tensor rank over $\mathbb{C}$ and $\mathbb{R}$) and it is numerically unstable (a tensor may be approximated arbitrarily well by tensors with strictly smaller rank, see eq. (17) below). In quantum information theory, the tensor rank appeared as a quantifier of entanglement, called the Schmidt measure [65], and as a tool to characterize entanglement transformations [66] and correlations [63].

The geometric measure for mixed states. – Up to this point, we considered the geometric measure $E(\psi)$ as a function of pure states only, but in practice, mixed quantum states are unavoidable. To extend the definition of geometric measure to mixed states, one needs to specify the required properties of an entanglement measure [3,4]. Standard conditions for a measure are that $E(\rho)$ should be positive on all states, *i.e.*, $E(\rho) \geq 0$, and $E(\rho) = 0$ should hold if ρ is separable. Additionally, $E(\rho)$ should not increase under positive maps that can be implemented by LOCC, so for any LOCC map L one has $E(\rho) \geq E(L(\rho))$. This implies invariance of E under local unitary transformations (which are reversible LOCC maps); but the class of general LOCC maps is hard to characterize.

One possibility of extending a potential entanglement measure from pure states to mixed states is given by the convex roof construction. In this construction one considers all convex decompositions $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ of the state ρ and takes the minimum

$$E(\rho) = \min_{p_i, |\psi_i\rangle} \sum_i p_i E(\psi_i) \quad (16)$$

over all decompositions. Clearly, this minimization is hard to evaluate, and a full solution of this problem has only been achieved for specific entanglement measures, such

as the two-qubit entanglement of formation [67] or some versions of the concurrence [68].

Typically, the convex roof construction inherits nice properties from $E(\psi)$, and for the geometric measure, it was shown that the convex roof $E(\rho)$ obeys all desired conditions for entanglement measures [7]. Although it is not clear from the construction at all, the convex-roof extended geometric measure has again a geometrical meaning [69]. If one measures the overlap (or inverse distance) of mixed states ρ and σ via the Uhlmann fidelity $F(\rho, \sigma) = [\text{Tr}(\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}})]^2$, then the geometric measure of mixed states quantifies the distance to the closest mixed fully separable state, via $E(\rho) = 1 - \max_{\sigma \in \text{f.s.}} F(\rho, \sigma)$.

There are some cases where the geometric measure of mixed states can be analytically determined (*e.g.*, for some specific bound entangled states [70]), or at least bounded from below [71,72]. For two qubits, $E(\rho)$ is related to the mixed state concurrence $C(\rho)$ [67] via $E(\rho) = \frac{1}{2}[1 - \sqrt{1 - C(\rho)^2}]$ [7,73]. However, this is not easy to generalize to the multipartite case or beyond qubits.

One can also obtain lower bounds on the geometric measure of an experimentally given state from incomplete measurement data, for example from measuring an entanglement witness $\mathcal{W}$ or a pure state fidelity $F_\phi = \langle\phi|\rho|\phi\rangle$ [9,74]. For computing $\eta = \inf\{E(\rho) \mid \text{Tr}(\mathcal{W}\rho) = w\}$ one can use the theory of the Legendre transform, and for the geometric measure, this can be evaluated with a see-saw algorithm [9] or even analytically, *e.g.*, if $\mathcal{W}$ is a fidelity measurement [2]. This has been used to calculate the geometric measure of mixed GHZ symmetric multi-qubit states or Dicke states affected by noise [75,76].

Discussion and extensions. – The geometric measure of entanglement combines physical relevance with mathematical elegance. It provides a playground where physical intuition as well as physical concepts (such as symmetries, separability criteria, and relaxations) help to gain insights into mathematical structures. Moreover, other problems such as the characterization of quantum channels can be related to the geometric measure, allowing to draw connections between different problems.

For future research, there are many interesting directions possible, here we would like to mention only two. First, while the quantity $\Lambda(\psi)$ characterizes the best approximation of $|\psi\rangle$ with product states, one may consider higher-order approximations, *e.g.*, by states of tensor rank two,

$$\Omega(\psi) = \sup_{|\phi\rangle = |abc\rangle + |def\rangle} |\langle\phi|\psi\rangle|. \quad (17)$$

Note that this is a so-called ill-posed problem, meaning that one has to consider the supremum instead of the maximum here [63,77], as the maximum may not be attained. This problem can be illustrated with the W state from eq. (7), which has tensor rank three but $\Omega(W) = 1$, since it can be approximated arbitrarily well by rank two states of the form $|\phi\rangle_\varepsilon \propto (|0\rangle + \varepsilon|1\rangle)^{\otimes 3} - |0\rangle^{\otimes 3}$ in the limit $\varepsilon \rightarrow 0$.

Other physically relevant approximations are pure stabilizer states or states from low-dimensional manifolds of the state space, but for these cases not much is known.

Second, it would be highly desirable to consider the approximation $\Lambda(\psi)$ from eq. (5) also with other normalization constraints of $|a\rangle$, $|b\rangle$, and $|c\rangle$. Examples are additional linear constraints of the type $\langle a|A|a\rangle = \alpha$. This may help to discuss optimizations over process tensors [78]; moreover, optimizations like eq. (17) may be reduced to eq. (5) with such constraints. In this way, the geometric measure will help to open further perspectives on various problems in quantum information theory.

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